A Calibrated Matched-Filter Framework for Deep and Rare Photometric Anomalies in High-Cadence Space Missions

Zane A. Graper
Independent Researcher, Fort Branch, Indiana, United States
zgraper@gmail.com
ORCID: 0009-0004-3810-6545

Revised author draft — 5 November 2026; not peer reviewed

Abstract

A fixed-duration matched filter can identify isolated dips in stellar brightness without requiring a repeating transit. This study presents a TESS photometry pipeline that combines a 2-hour box filter, empirical threshold calibration, continuum-referenced verification, and streaming archive access. A threshold of $T = 73.7797$ targeted a 1.0% development alert-window fraction. Independent evaluation on 4,000 stars gave alert-window rates of $1.144\% \pm 0.211\%$ in Sectors 1–10 and $2.184\% \pm 0.302\%$ in Sector 11; these rates are not catalog false-positive probabilities. In specified synthetic tests, the original verifier rejected 47/47 flare-induced apparent dips and retained 161/161 threshold-triggered injections when evaluated at their known geometry. Trigger fractions for 0.5%-deep, 4-hour and 8-hour injections were 6.2% and 0.0%, showing a limitation of the fixed-duration filter. Historical classifications are retained as original-verifier outputs and do not establish performance for the corrected `coherence_v2` gate. Independent expert adjudication remains pending.

Keywords: methods: data analysis — techniques: photometric — stars: variables: general

1 Introduction

A single dip in stellar brightness can be scientifically useful even when it does not repeat during an observing sector. Possible causes include a long-period transit, an eclipse, or obscuration by circumstellar material. The same light-curve shape can also arise from instrumental effects or changes in the reference brightness used by a detector. A useful search must therefore distinguish an unusual measurement from an established physical explanation.

NASA's Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2015) provides high-cadence photometry for such searches. Periodic methods, including Box Least Squares (BLS; Kovács et al. 2002) and Transit Least Squares (TLS; Hippke and Heller 2019), combine evidence across repeated transits. Single-event searches instead examine local changes in brightness and require separate control of false alerts.

This study examines a fixed 2-hour matched filter for deep, localized dimmings. The method compares a candidate interval with neighboring reference intervals, called sidebands. Its main vulnerability is simple: a flare in a sideband can raise the reference flux and make an unchanged central interval appear dimmer. A continuum-referenced verifier checks that apparent dip against a broader local baseline. The paper reports historical development and independent validation results, with explicit limits on what the detector, verifier, and subsequent classifications establish.

1.1 Related approaches

Single-transit searches have been developed for Kepler, K2, and TESS [Foreman-Mackey et al., 2016, Osborn et al., 2016, Cooke et al., 2018]. Other approaches use machine learning to

identify unusual light curves [Giles and Walkowicz, 2019, Pearson et al., 2018]. Automated vetting systems, including the Kepler Robovetter and DAVE, assess competing explanations for candidate events [Coughlin et al., 2016, Kostov et al., 2019]. The present study focuses on a transparent local score, an empirically chosen operating threshold, and a documented separation between photometric classification and physical attribution. It does not establish that these components are absent from other methods.

2 Detection and verification

2.1 Fixed-duration score

The pipeline uses TESS SPOC light curves sampled every 2 minutes. Analysis windows contain 512 cadences and advance by 256 cadences, corresponding to nominal coverages of 17.07 hours and 8.53 hours. These intervals are distinct from the 2-hour filter duration and the approximately 27-day sector baseline. Quality masks and gaps are retained without interpolation.

Let $f_i = f(t_i)$ denote normalized flux at a valid cadence. The noise scale is estimated from adjacent valid-cadence differences, using pairs with $\Delta t_i < 1.55$ times the nominal cadence:

$$\sigma = \frac{1.4826}{\sqrt{2}} \,\mathrm{median}\left(|\Delta f_i - \mathrm{median}(\Delta f)|\right). \tag{1}$$

This estimate describes short-timescale scatter; it does not remove correlated variability.

The core interval C contains approximately $W = 60$ samples for a 2-hour template. Each sideband contains

$$W_{\mathrm{side}} = \min\{60, \max(8, W \,//\, 2), \max[8, (N - W) \,//\, 4]\}, \tag{2}$$

where N is the analysis-window length and // denotes integer division. At the nominal cadence, the core and two 1-hour sidebands span approximately 4 hours. Search centers advance by approximately 6 samples (12 minutes). With N_{core} valid core samples, the detector computes

$$\bar{f}_{\mathrm{base}} = \tfrac{1}{2}(\bar{f}_L + \bar{f}_R), \tag{3}$$

$$\Delta f = \bar{f}_{\mathrm{base}} - \bar{f}_{\mathrm{core}}, \tag{4}$$

$$S = \frac{\Delta f \sqrt{N_{\mathrm{core}}}}{\sigma}. \tag{5}$$

The score is set to zero when $\Delta f \leq 0$. The quantity Δf is a sideband-referenced decrement, not a fitted eclipse depth. A large score is evidence of contrast under this filter, not a calibrated Gaussian detection significance.

2.2 Flare-induced apparent dips

If one sideband is raised by mean flare flux $\Delta f_{\mathrm{flare}}$ while the core remains at the continuum, the apparent decrement is $\Delta f = \frac{1}{2}\Delta f_{\mathrm{flare}}$. The resulting score is

$$S = \frac{\Delta f_{\mathrm{flare}} \sqrt{N_{\mathrm{core}}}}{2\sigma}. \tag{6}$$

For $\sigma = 350$ ppm, $N_{\mathrm{core}} \simeq 60$, and a 1.0% sideband elevation, $S \simeq 110.7$, above the operating threshold of 73.78. The detector can therefore trigger on a quiet core next to a flare (Figure 2).

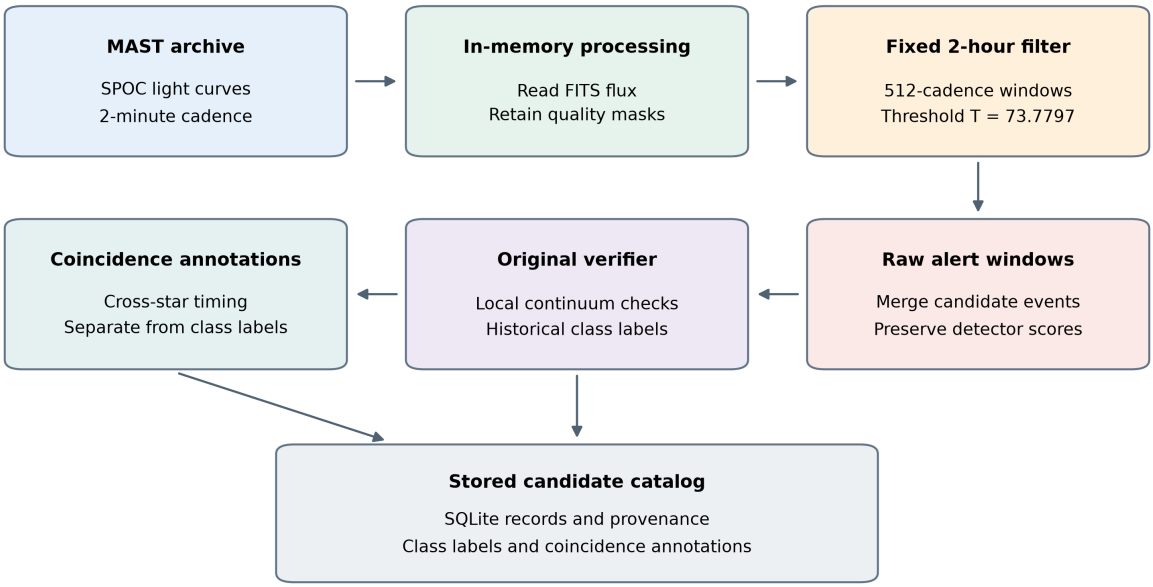


Figure 1: Historical survey workflow. Archive flux is scored in memory; candidate records and results are stored on disk. Photometric verification and cross-star coincidence annotations address different questions. The historical classifier shown here is the original implementation, not the later `coherence_v2` gate.

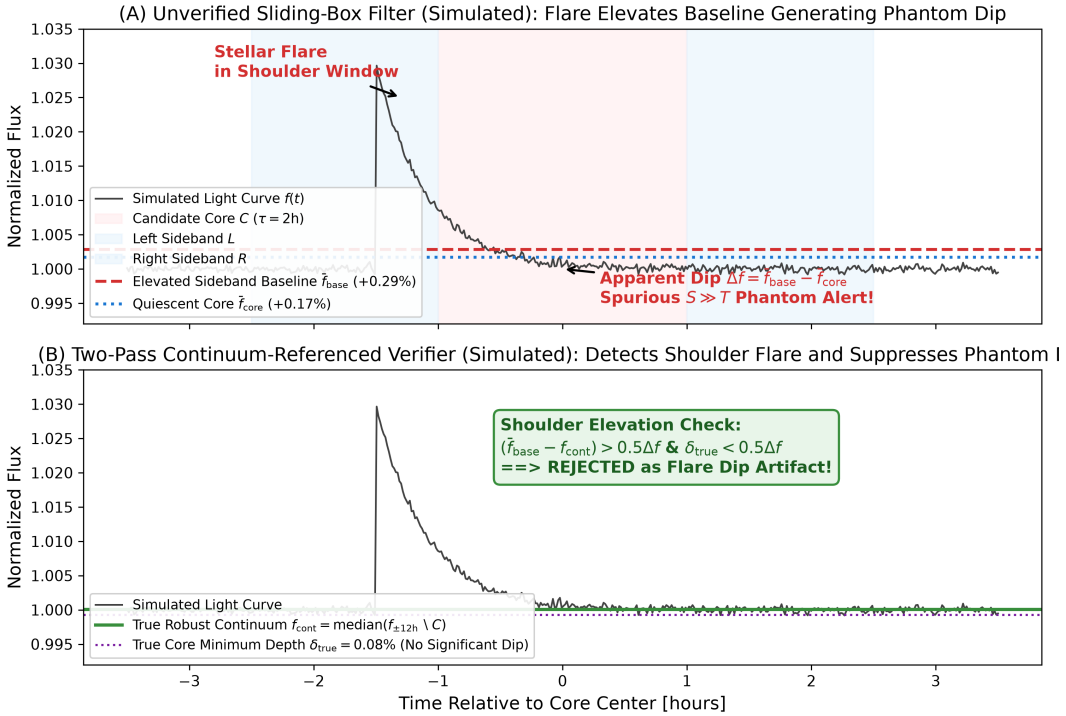


Figure 2: Simulated illustration of sideband contamination. A flare raises the sideband reference in the upper panel. The broader continuum in the lower panel provides a separate check on whether the core is actually depressed. This illustration explains the geometry; it is not an empirical validation experiment.

2.3 Original verifier and subsequent correction

The original verifier estimated a local continuum from good data within ±12 hours, excluded the region within one candidate width of the center, and clipped positive excursions above 3σ, with fallback and noise-floor rules. It measured both the core-mean decrement and the continuum-referenced minimum-sample decrement,

$$\delta_{\min} = f_{\text{cont}} - \min_{i \in C} f_i. \tag{7}$$

The latter is sensitive to a single low sample and should not be interpreted as a model-fitted transit depth. The shoulder-elevation check identified flare-like apparent dips when $(\bar{f}_{\text{base}} - f_{\text{cont}}) > 0.5\Delta f$ and $\delta_{\min} < 0.5\Delta f$.

The intended coherence criterion was at least two consecutive samples below $f_{\text{cont}} - 1.5\sigma_{\text{local}}$. A subsequent software assessment found that the historical implementation did not guarantee timestamp adjacency: separated low samples and gaps in retained data could satisfy its checks. The public software now has a separately versioned `coherence_v2` gate that checks adjacent good cadences in time. All classifications and synthetic retention results reported here belong to the original verifier. No event-level reclassification or population benchmark for the corrected gate is reported.

Historical labels distinguish localized dimmings, continuous variability, flare-induced apparent dips, probable instrumental artifacts, and inconclusive cases. These are algorithmic descriptions. A "verified localized dimming" label does not establish its source or guarantee the corrected coherence criterion.

2.4 Empirical calibration

The development calibration used 4,043 unique stars and 236,130 windows. A pooled empirical quantile for the target alert-window fraction $\alpha = 0.010$ gave

$$T = 73.77965536097444 \qquad (\texttt{fixed_box_2h}). \tag{8}$$

The reported development cluster-ratio standard error was 0.187%, using star-level block bootstrap resampling with $B = 2,000$ [Wolter, 2007]. Overlapping windows are dependent. This threshold targets an alert-window fraction under the calibration distribution, not a universal catalog false-alarm probability.

3 Streaming and cross-star coincidence

The streaming engine queries MAST, reads FITS data into memory, scores analysis windows, writes candidate records to SQLite, and releases temporary flux buffers. Processing archive flux in memory reduces local FITS storage; catalogs and other outputs still persist on disk. The historical validation run used four concurrent workers and processed 4,000 observations comprising 59,771,463 usable cadences with no reported execution failures. Its logs summarized 36.69 minutes and 6,364.9 observations per hour, but those two values imply different timing denominators. They remain unreconciled, so no precise throughput benchmark is claimed here.

Cross-star coincidence is assessed separately from photometric shape. A gap-aware Monte Carlo null model with 5,000 simulations retained sector duration, inter-orbit gaps, and star-level event frequencies. It used sector-wide coverage rather than individual target masks. For exactly three cameras, the reported p value was 0.0736; this is not a cumulative result for three or more cameras. Across 14 multiplicity thresholds ($k \geq 3$ through $k \geq 16$), Benjamini–Hochberg control at $q = 0.05$ [Benjamini and Hochberg, 1995] selected $k \geq 5$, $k \geq 6$, and $k \geq 16$, with respective p values 0.0010, 0.0092, and 0.0064. These population tests do not establish a spacecraft cause or provide calibrated event-level probabilities for every catalog risk flag.

4 Historical development survey

The exploratory survey covered 9,753 observations of 8,984 stars in Sectors 1–10. It produced 4,827 merged events across 305 stars. The original verifier labeled 2,435 events as localized dimmings, 2,213 as continuous variability, 110 as flare-induced apparent dips, and 69 as probable instrumental artifacts. Of the localized-dimming labels, 815 had no detected cross-star coincidence under the specified search; that absence is not proof of an astrophysical origin.

The historical benchmark set included EBLM J0048-66 (TIC 39018208), Y Hyi (TIC 262958558), and ST Tau (TIC 247585034). The recorded BLS period for EBLM J0048-66 was 6.6487 days, close to the cited orbital period of 6.64927 days [Triaud et al., 2017]. The original diagnostic and its limitations are retained in Appendix D. The analysis used phase-bin minima rather than an eclipse-profile fit. It therefore does not establish the previously quoted 0.482% fitted secondary depth, 8.91σ detection, or quantitative eccentricity constraint. Those claims are withdrawn in this revision pending a model fit with uncertainty and selection accounting. The public same-TIC demonstration is synthetic and is not evidence for this system. Other historical benchmark entries recorded Y Hyi at $P = 3.53597$ days with a 78.93% decrement, and ST Tau at $P = 4.06757$ days compared with the quoted literature value of 4.0343 days, with a 41.67% decrement. These are inherited catalog summaries, not new model fits.

5 Independent validation

5.1 Cohorts and alert-window rates

The independent study used 4,000 previously unexamined stars: 2,000 in Sectors 1–10 and 2,000 in Sector 11. A target-overlap check recorded no shared stars with the 8,984-star development universe. The in-sector cohort tests transfer to new stars in previously used sectors; Sector 11 tests transfer to one unseen sector. Neither cohort alone establishes mission-wide performance.

Standard errors were estimated with star-clustered Taylor linearization, with bootstrap intervals from 2,000 star-level resamples [Wolter, 2007]. Table 1 summarizes the historical results. The in-sector alert-window rate was $1.144\% \pm 0.211\%$, compared with the nominal 1.0% target ($z_{\mathrm{cal}} = 0.69$, $p = 0.49$). Failure to reject this target is not evidence of equivalence. Sector 11 had an elevated rate of $2.184\% \pm 0.302\%$ ($z_{\mathrm{cal}} = 3.92$, $p < 0.001$).

Table 1: Historical independent validation at $T = 73.7797$. Uncertainties are star-clustered standard errors. Class counts are original-verifier outputs.

Metric	Sectors 1–10	Sector 11	Combined
Stars	2,000	2,000	4,000
Evaluated windows	109,229	97,339	206,568
Raw alert windows	1,250	2,126	3,376
Alert-window rate	1.144%	2.184%	1.634%
Standard error	0.211%	0.302%	0.181%
Bootstrap 95% interval	[0.736, 1.578]%	[1.622, 2.794]%	[1.270, 2.014]%
Stars with raw alerts	61	88	149
Merged candidate events	903	1,616	2,519
Localized dimming labels	449	583	1,032
Continuous variability labels	414	1,006	1,420
Flare-induced apparent dips	24	20	44
Probable instrumental artifacts	16	7	23
Inconclusive	0	0	0
Stars with localized dimming labels	35	52	87

Sector rates ranged from 0.036% in Sector 10 to 2.646% in Sector 9 (Figure 3; Appendix A).

In Sector 11, continuous modulation accounted for 1,006/1,616 merged events (62.3%), while 7/1,616 (0.4%) received probable-artifact labels. These labels do not measure instrumental contamination or establish why the sector rate increased. Stellar variability, field composition, and instrumental background are possible contributors requiring separate tests.

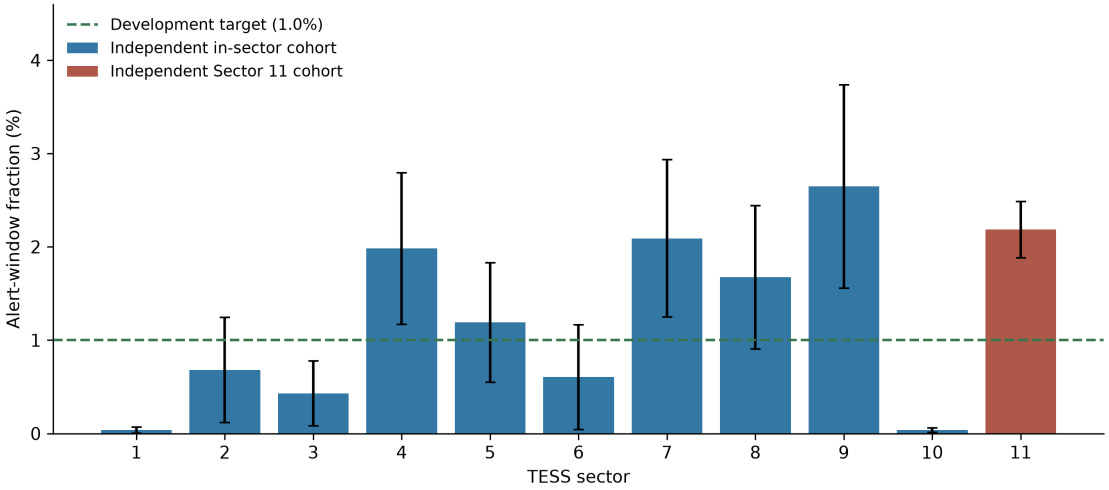


Figure 3: Historical validation alert-window fractions by sector. Error bars show star-clustered standard errors; the dashed line marks the 1.0% development target. Colors identify independent validation cohorts, not inferred physical causes of the rate differences. Values are listed in Appendix A.

5.2 Synthetic tests and duration mismatch

The flare experiment used 50 simulated Gaussian-noise baselines, not 50 observed quiet stars. Injected fast-rise, exponential-decay flares had amplitudes of 1%–10% and decay times of 15–60 minutes [Davenport, 2016]; baseline scatter was 300–450 ppm. The detector triggered on 47/50 trials, and the original verifier rejected all 47 resulting apparent dips. This is an observed proportion in a specified finite experiment, not universal flare suppression.

Across 192 transit-injection trials, 161 windows crossed threshold and all 161 were retained by the original verifier. Verification used the known injection center and width; detection used the fixed 2-hour filter and did not require recovery at the correct event time. These experiments therefore measure window-trigger recovery followed by verification at known geometry, rather than complete operational event recovery. The full grid is in Appendix A.

For 0.5%-deep injections, trigger fractions were 16/16 for 2-hour events, 1/16 (6.2%) for 4-hour events, and 0/16 for 8-hour events. All tested deeper cells triggered in 16/16 trials. These finite samples show duration mismatch under the specified simulation; 100% in one cell is not a theoretical sensitivity bound. They also do not validate `coherence_v2`.

5.3 Expert adjudication remains pending

The prepared review sample contains 300 blinded review items: 100 events from the 1,032 localized-dimming labels, 100 from the 1,487 other candidate events, and 100 stars from 3,851 zero-event controls. Event and star sampling frames are distinct. Review materials omit detector scores, internal labels, and stratum assignments, and use randomized identifiers with a separate decoding key. Reviewers are asked to assess the observed flux excursion separately from its possible physical cause. Recruitment and ratings remain pending; no expert precision, missed-event prevalence, or consensus result is reported. Design weights and estimators are given in Appendix B.

6 Discussion and conclusions

The fixed-duration score provides an interpretable way to screen isolated dimmings, but its output depends on the reference intervals and the observing population. A sideband flare can create a strong score without a central flux deficit. Comparing the event with a broader continuum addresses this mechanism, although the original verifier's adjacency limitation prevents treating its labels as guarantees of time-coherent events.

The independent cohorts show that empirical alert-window fractions change across fields. The pooled in-sector result did not significantly differ from the development target, whereas Sector 11 did. The appropriate conclusion is limited transfer of one operating threshold, with substantial sector variation. A raw alert fraction cannot replace expert classification, source attribution, or a catalog contamination estimate.

A fixed 2-hour filter also loses sensitivity to some broader shallow events. A bank of durations could improve coverage [Osborn et al., 2016], but would require new calibration of the combined search. The present threshold does not automatically apply to other widths. Real stellar baselines introduce additional variation and correlated noise [Pont et al., 2006]; the legacy real-baseline comparison is retained only as supporting context in Appendix C.

The main results are the empirically calibrated detector, the demonstrated sideband-contamination mechanism, and the historical alert-rate evaluation on star-disjoint cohorts. Stronger claims require operational event-recovery tests for the corrected verifier, independent expert review, and model-based analysis of individual systems. Cross-star coincidence also needs caution because the current null model lacks per-target cadence masks. TESS pixels span approximately $21''$, so source attribution may require pixel-level checks, higher-resolution imaging, and independent observations.

Data and software availability

TESS light curves are available from MAST. The public MIT-licensed software beta is hosted at `https://github.com/zgraper/TESS_Anomaly_Survey`; its package version is `0.3.0b2`. The public distribution includes software and synthetic demonstration data, but not every historical calibration input, research database, or adjudication record. Prepared supplementary packages should not be assumed publicly accessible unless a specific archive or release is identified. The adjudication decoding key remains private to preserve blinding.

Frozen source hashes, configuration records, and seeds document implementation identity and aspects of provenance. They do not by themselves make all historical results reproducible. A complete reconstruction still requires the exact archive products, masks, cohort definitions, original verifier version, analysis environment, and figure inputs. This revision changes presentation and interpretation; it does not rerun the survey or generate replacement scientific results.

Disclosure of artificial intelligence use

OpenAI ChatGPT and Google Gemini were used as assistive tools across the development of the research, software, and manuscript, including research planning, analysis support, code development and debugging, and drafting and revision of text. Their contributions were iterative and are disclosed collectively rather than assigned to individual passages or code components. These systems are not authors. Responsibility for the study's methods, evidence, citations, software, interpretation, and final manuscript rests with the human author. AI assistance does not substitute for independent validation or expert review; the pending checks and limitations described in this paper remain applicable.

A Detailed validation and injection tables

The following tables preserve historical numerical outputs. Classifications use the original verifier, and injection retention uses known injection geometry. No results have been relabeled as outputs of the corrected gate.

Table 2: Historical validation results by sector. Verified and variable counts are original-verifier labels, not physical confirmations or outputs of `coherence_v2`.

Sector	Cohort	Stars	Windows	Alerts	Alert Rate	Cluster SE	Verified	Variable
1	In-Sector	259	16,037	6	0.037%	±0.032%	0	0
2	In-Sector	181	9,533	65	0.682%	±0.564%	6	42
3	In-Sector	157	6,985	30	0.429%	±0.347%	16	0
4	In-Sector	235	11,052	219	1.982%	±0.812%	103	66
5	In-Sector	200	12,016	143	1.190%	±0.641%	91	3
6	In-Sector	179	8,934	54	0.604%	±0.561%	3	40
7	In-Sector	196	12,392	259	2.090%	±0.843%	88	75
8	In-Sector	203	9,496	159	1.674%	±0.768%	46	63
9	In-Sector	196	11,753	311	2.646%	±1.090%	96	125
10	In-Sector	194	11,031	4	0.036%	±0.026%	0	0
11	Cross-Sector	2,000	97,339	2,126	2.184%	±0.302%	583	1,006

Table 3: Historical synthetic injection grid for the fixed 2-hour detector. Each cell contains 16 trials. Detection means a window score exceeded threshold; original-verifier retention was evaluated at the known injection geometry, not the detected event geometry.

Duration	Depth	Trials	Triggers	Retained	Trigger %	Retention %
2.0 h	0.5%	16	16	16	**100.0%**	**100.0%**
2.0 h	1.0%	16	16	16	**100.0%**	**100.0%**
2.0 h	2.0%	16	16	16	**100.0%**	**100.0%**
2.0 h	5.0%	16	16	16	**100.0%**	**100.0%**
4.0 h	0.5%	16	1	1	**6.2%**	**100.0%**
4.0 h	1.0%	16	16	16	**100.0%**	**100.0%**
4.0 h	2.0%	16	16	16	**100.0%**	**100.0%**
4.0 h	5.0%	16	16	16	**100.0%**	**100.0%**
8.0 h	0.5%	16	0	0	**0.0%**	N/A
8.0 h	1.0%	16	16	16	**100.0%**	**100.0%**
8.0 h	2.0%	16	16	16	**100.0%**	**100.0%**
8.0 h	5.0%	16	16	16	**100.0%**	**100.0%**

B Adjudication design and estimators

The event frame contains $N_1 = 1,032$ localized-dimming labels and $N_2 = 1,487$ other candidates, with $n_1 = n_2 = 100$ sampled items. The corresponding design weights are $w_1 = 10.3200$ and $w_2 = 14.8700$. For a binary expert outcome y_{hi} in stratum h, the weighted fraction of reviewed outcomes over the full candidate frame is

$$\hat{P}_{\text{event}} = \frac{w_1 \sum_{i=1}^{n_1} y_{1i} + w_2 \sum_{i=1}^{n_2} y_{2i}}{N_1 + N_2}. \tag{9}$$

This estimates the fraction satisfying the chosen expert criterion among all candidates; it is not automatically precision restricted to the localized-dimming class. Under stratified simple

random sampling, the planned variance estimator is

$$\widehat{\mathrm{Var}}(\hat{P}_{\mathrm{event}}) = \frac{1}{(N_1+N_2)^2}\sum_{h=1}^{2} N_h^2\left(1-\frac{n_h}{N_h}\right)\frac{s_h^2}{n_h}, \tag{10}$$

where s_h^2 is the within-stratum sample variance [Horvitz and Thompson, 1952]. Any dependence from multiple events on a star should be considered when interpreting uncertainty.

The negative-control frame contains $N_3 = 3,851$ zero-event stars, with $n_3 = 100$ sampled stars and $w_3 = 38.5100$. If y_{3i} indicates an expert-identified missed event in a full light curve, the planned star-level prevalence and variance estimators are

$$\hat{P}_{\mathrm{miss,star}} = \frac{1}{n_3}\sum_{i=1}^{n_3} y_{3i}, \qquad \widehat{\mathrm{Var}}(\hat{P}_{\mathrm{miss,star}}) = \left(1-\frac{n_3}{N_3}\right)\frac{s_3^2}{n_3}. \tag{11}$$

This is a star-level missed-event prevalence, not event-level recall. No reviewer outcomes have yet been supplied.

C Legacy real-baseline recovery comparison

The historical figure source records 33.3% recovery for 2-hour, 0.5%-deep injections in the Phase 6A real-baseline experiment, compared with 16/16 triggers in the Phase 6B Gaussian-noise cell. The original two-panel display is retained as a comparison of historical summaries (Figure 4). Its real-baseline series is inherited from hardcoded figure-source values and still requires event-level provenance reconciliation. It is not evidence that noise alone caused the difference. A nominal calculation gives $5000\sqrt{60}/300 \simeq 129.1$ for 300 ppm scatter and $5000\sqrt{60}/1000 \simeq 38.7$ for an assumed 1,000 ppm scale; this only illustrates dependence on the chosen noise scale. A single-target red-noise factor cannot be applied to the full injection cohort.

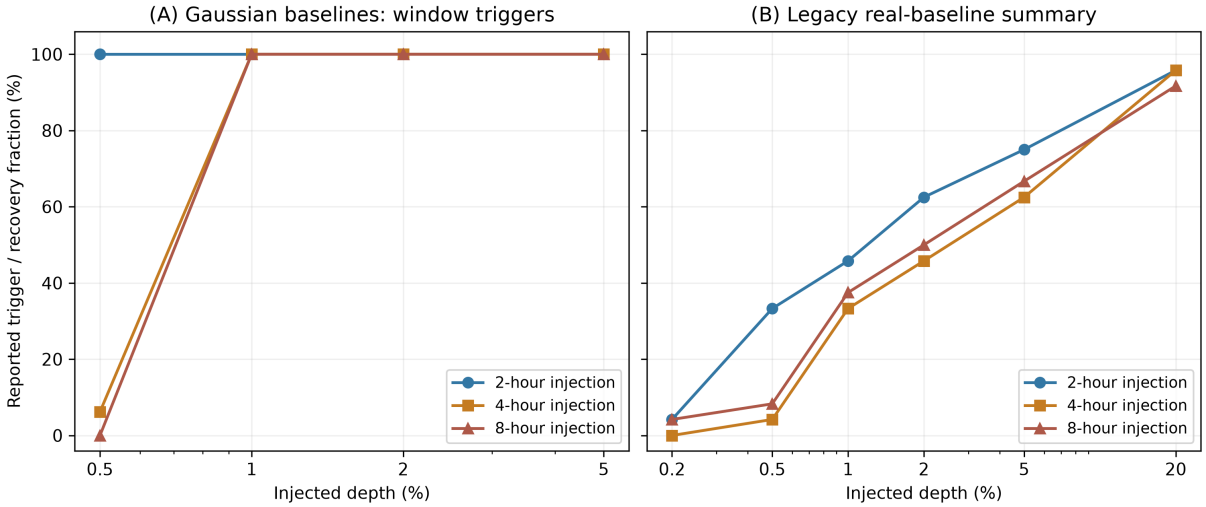


Figure 4: Historical injection summaries redrawn without causal annotations. Left: Gaussian-baseline window triggers (16 trials per cell), followed by original verification at known geometry. Right: legacy Phase 6A real-baseline values inherited from the original figure source; event-level provenance remains to be reconciled. The panels do not establish equivalent operational recovery procedures or causal noise effects.

D Legacy EBLM diagnostic

Figure 5 preserves the earlier diagnostic rather than presenting a new fit. Its displayed period is 6.655331 days, with primary and secondary annotations of approximately 2.97% and 0.72%; these differ from the phase-bin measurements quoted elsewhere in the original draft. Neither the annotations nor the phase-bin minima establish a fitted secondary depth or its uncertainty. A revised physical analysis must reconcile ephemerides, flux products, detrending, individual eclipses, correlated noise, and selection of phase bins before reporting detection significance or an eccentricity constraint.

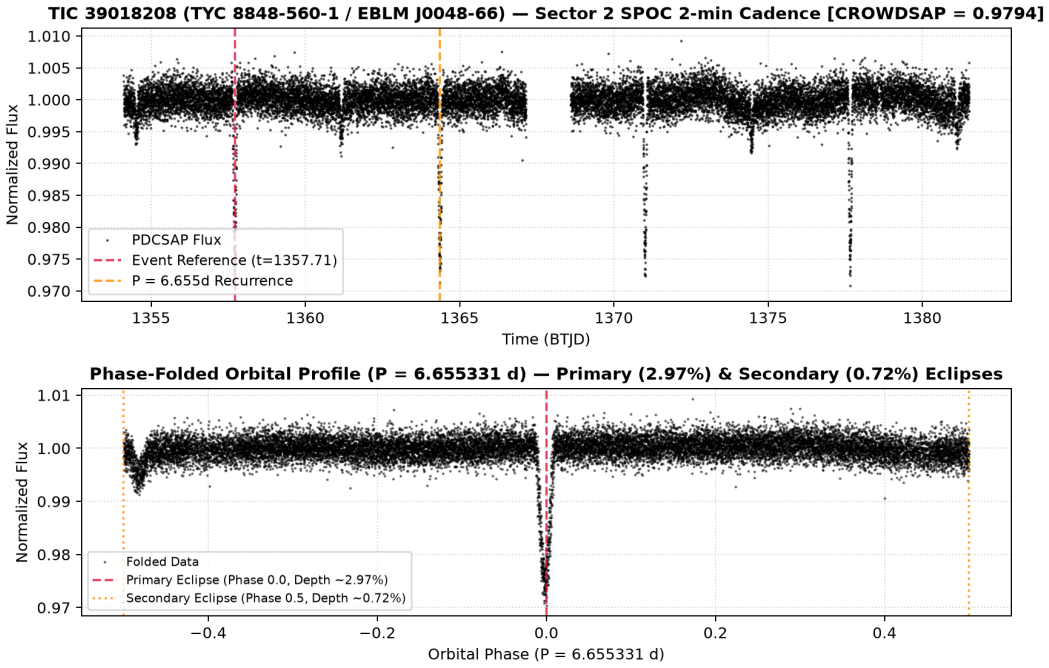


Figure 5: Unmodified legacy diagnostic associated with TIC 39018208, Sector 2. The upper panel is the time series and the lower panel is a folded light curve. Annotations are historical exploratory measurements, not results of a fitted eclipse model. This figure is retained for provenance, with discrepancies stated in the text.

References

Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

B. F. Cooke, D. Pollacco, R. West, J. McCormac, and P. J. Wheatley. Single site observations of TESS single transit detections. *Astronomy & Astrophysics*, 619:A175, 2018. doi: 10.1051/0004-6361/201834014.

J. L. Coughlin, F. Mullally, S. E. Thompson, J. F. Rowe, C. J. Burke, D. W. Latham, N. M. Batalha, and S. T. Bryson. Planetary candidates observed by Kepler. VII. the first fully uniform catalog based on the entire 48-month data set (Q1–Q17 DR24). *The Astrophysical Journal Supplement Series*, 224(1):12, 2016. doi: 10.3847/0067-0049/224/1/12.

J. R. A. Davenport. The Kepler catalog of stellar flares. *The Astrophysical Journal*, 829(1):23, 2016. doi: 10.3847/0004-637X/829/1/23.

D. Foreman-Mackey, T. D. Morton, D. W. Hogg, E. Agol, and B. Schölkopf. The population of long-period transiting exoplanets. *The Astronomical Journal*, 152(6):206, 2016. doi: 10.3847/0004-6256/152/6/206.

D. Giles and L. Walkowicz. Systematic serendipity: a test of unsupervised machine learning as a method for anomaly detection. *Monthly Notices of the Royal Astronomical Society*, 484(1): 834–849, 2019. doi: 10.1093/mnras/sty3461.

M. Hippke and R. Heller. Optimized transit detection algorithm to search for periodic transits of small planets. *Astronomy & Astrophysics*, 623:A39, 2019. doi: 10.1051/0004-6361/201834672.

D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952. doi: 10.1080/01621459.1952.10483446.

V. B. Kostov, S. E. Mullally, E. V. Quintana, J. L. Coughlin, F. Mullally, T. Barclay, K. D. Colon, and J. E. Schlieder. Discovery and vetting of exoplanets. I. benchmarking K2 vetting tools. *The Astronomical Journal*, 157(3):124, 2019. doi: 10.3847/1538-3881/ab0110.

G. Kovács, S. Zucker, and T. Mazeh. A box-fitting algorithm in the search for periodic transits. *Astronomy & Astrophysics*, 391(1):369–377, 2002. doi: 10.1051/0004-6361:20020802.

H. P. Osborn, D. J. Armstrong, D. J. A. Brown, J. McCormac, A. P. Doyle, T. Louden, J. Kirk, and J. J. Spake. Single transit candidates from K2: detection and period estimation. *Monthly Notices of the Royal Astronomical Society*, 457(3):2273–2286, 2016. doi: 10.1093/mnras/stw137.

K. A. Pearson, L. Palafox, and C. A. Griffith. Searching for exoplanets using artificial intelligence. *Monthly Notices of the Royal Astronomical Society*, 474(1):478–491, 2018. doi: 10.1093/mnras/stx2761.

F. Pont, S. Zucker, and D. Queloz. The effect of red noise on planetary transit detection. *Monthly Notices of the Royal Astronomical Society*, 373(1):231–242, 2006. doi: 10.1111/j.1365-2966.2006.11012.x.

G. R. Ricker, J. N. Winn, R. Vanderspek, D. W. Latham, G. Á. Bakos, J. L. Bean, Z. K. Berta-Thompson, and T. M. Brown. Transiting exoplanet survey satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1):014003, 2015. doi: 10.1117/1.JATIS.1.1.014003.

A. H. M. J. Triaud, D. V. Martin, D. Ségransan, B. Smalley, P. F. L. Maxted, D. R. Anderson, F. Bouchy, and A. Collier Cameron. The EBLM project. IV. spectroscopic orbits of over 100 eclipsing M dwarfs orbiting F/G/K primaries. *Astronomy & Astrophysics*, 608:A129, 2017. doi: 10.1051/0004-6361/201730993.

K. M. Wolter. *Introduction to Variance Estimation*. Springer, New York, 2nd edition, 2007. doi: 10.1007/978-0-387-35099-8.